Git & GitHub — Basics

Internship Day 3 Task

1. What is Git?

Git is a distributed version control system. It tracks changes to files over time, so a project's history can be reviewed, compared, and reverted whenever needed. It runs entirely on the local machine, which means commits, branches, and history are all available offline.

2. What is GitHub?

GitHub is a cloud platform that hosts Git repositories online. It adds collaboration features on top of Git — pull requests, issue tracking, code review, and a shared remote copy of the project that a team can push to and pull from.

In short: Git is the tool, GitHub is the hosting service built around it.

3. Core Concepts

- Repository (repo): a project folder tracked by Git.
- Commit: a saved snapshot of changes, with a message describing them.
- Branch: an independent line of development, used to build features without disturbing the main codebase.
- Remote: a version of the repository hosted elsewhere, such as on GitHub.
- Clone: a local copy of a remote repository.
- Pull request (PR): a request to merge changes from one branch into another, reviewed on GitHub before merging.

4. Setting Up Git (one-time)

```
git config --global user.name "Your Name"

git config --global user.email "you@example.com"

git config --list
```

5. Starting a Repository

Create a new repo locally:

```
git init
```

Or copy an existing one from GitHub:

```
git clone https://github.com/username/repo-name.git
```

6. The Daily Workflow

```
git status                 # see what has changed
git add file-name          # stage a specific file
git add .                  # stage everything changed
git commit -m "message"    # save the staged changes
git push                   # upload commits to GitHub
git pull                   # download latest changes from GitHub
```

7. Working with Branches

```
git branch                     # list branches
git branch feature-name        # create a new branch
git checkout feature-name      # switch to that branch
git checkout -b feature-name   # create and switch in one step
git merge feature-name         # merge a branch into the current one
git branch -d feature-name     # delete a branch after merging
```

8. Viewing History

```
git log                 # full commit history
git log --oneline       # condensed, one line per commit
git diff                # see unstaged changes
```

9. Connecting a Local Repo to GitHub

After creating an empty repository on GitHub, link it to a local project:

```
git remote add origin https://github.com/username/repo-name.git
git branch -M main
git push -u origin main
```

10. Undoing Changes

```
git checkout -- file-name     # discard local changes in a file
git reset file-name           # unstage a file
git revert commit-hash        # create a new commit that undoes an old one
```

11. Typical Flow for a Task Submission

- Clone or pull the latest repository.
- Create a branch for the task (optional for small tasks).
- Make changes and save the file.
- Stage and commit with a clear message.
- Push to GitHub.
- Copy the repository or file link and share it with the mentor.

12. Quick Reference

```
git init            git clone <url>
git status          git add <file>
git commit -m ""    git push
git pull            git branch
git checkout -b     git merge
git log --oneline   git diff
```